Microsoft AI Engineer Program via Simplilearn

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Mercedes-Benz Greener Manufacturing

Course-end Project 1 — Writeup

Problem

Mercedes-Benz runs every unique car configuration through a physical test bench before it's cleared for the road. That's slow, and with the number of possible option combinations on a modern car, it doesn't scale. The goal here is a model that predicts test bench time (`y`) from a configuration's features, so testing can be prioritized instead of run blind.

Data

`train.csv` has 4,209 rows and 378 columns, including `y` . `test.csv` has 4,209 rows and 377 columns, no `y` . Each row is one anonymized configuration:

- `X0` - `X8` : categorical, letter-coded options
- `X10` - `X385` : binary flags, one per individual feature
- `y` : test bench time in seconds, train only

What I did

Zero-variance columns. Ran `nunique()` on every feature besides `ID` and `y` . 12 columns came back with a single value across the whole training set: `X11, X93, X107, X233, X235, X268, X289, X290, X293, X297, X330, X347` . Dropped all 12 from both train and test — 376 features down to 364.

Nulls and uniques. No nulls anywhere in either file. Checked unique counts per column too, mostly to confirm which ones were categorical (`X0` - `X8`, tens of string values) versus binary (exactly 0/1).

Label encoding. `X0` through `X8` went through `LabelEncoder`, fit on train and test combined first so a category that only shows up in test doesn't break the transform.

PCA. 364 features left at this point, mostly binary and heavily correlated with each other. Ran PCA asking for 95% of the variance and it collapsed down to 6 components. Fit on train, applied to test.

XGBoost. 500 trees, depth 4, learning rate 0.05, 80% subsampling on both rows and columns. Held out 20% for validation first, then refit on the full training set before predicting on test.

Results

Metric	Value
Zero-variance columns removed	12
Nulls found (train / test)	0 / 0
Categorical columns label-encoded	8 (`X0` - `X8`)
Features before PCA	364
Components after PCA (95% variance)	6
Validation R^2	0.387
Validation RMSE	9.77 seconds

Wrote predictions for all 4,209 test rows to `mercedes_benz_test_predictions.csv` (`ID`, `y`).

Notes

R^2 of 0.39 isn't great on its own, but it lines up with what's known about this dataset. There are outlier configurations in the original data with `y` values well above the normal 70-150 second range, and those drag error up for a model that isn't treating them specially. All five required steps made it into the pipeline: zero-variance removal, null/unique checks, label encoding, PCA, XGBoost. Going from 364 features to 6 components is an aggressive cut, and it costs some accuracy, but the payoff is a much smaller and faster model. For a testing-time-reduction use case that seems like a fair trade, since prediction speed matters here almost as much as raw accuracy.